

NORFOLK INTL, VA, USA

WMO: 723080

Lat: 36.903N

Lon: 76.192W

Elev: 30

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	25.8	2.6	6.3	26.2	7.7	8.2	30.5	27.1	51.9	24.7	51.4	9.1	350	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	93.5	76.8	91.2	76.0	89.0	75.3	79.3	88.4	78.2	86.9	77.3	85.6	10.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	140.7	83.1	75.9	135.4	82.1	74.9	131.1	81.4	42.7	88.2	41.6	87.0	40.6	85.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.2	20.6	18.6	DB	16.9	98.3	5.0	2.3	13.3	99.9	10.4	101.3	7.5	102.6	3.9	104.2
			WB	14.1	81.3	4.5	1.5	10.8	82.4	8.2	83.2	5.6	84.0	2.3	85.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	41.5	43.9	50.2	59.8	67.9	76.3	80.1	78.7	73.5	63.5	52.8	46.1
	DBStd	15.57	10.30	9.41	9.34	8.55	7.34	5.73	4.32	4.05	5.02	7.39	8.18	9.13
	HDD50	900	304	218	117	12	0	0	0	0	0	2	60	187
	HDD65	3139	729	593	469	204	56	3	0	0	3	119	374	589
	CDD50	5017	41	46	121	307	555	788	934	891	705	419	144	66
	CDD65	1783	1	1	9	49	147	341	469	426	258	71	8	3
	CDH74	14916	2	12	73	399	1115	2975	4659	3735	1536	368	32	10
	CDH80	5138	0	0	7	102	345	1109	1863	1263	378	69	2	0
Wind	WSAvg	9.0	9.4	9.6	10.3	10.5	9.5	8.5	8.1	7.6	8.8	8.6	8.7	8.6
Precipitation	PrecAvg	45.4	3.7	3.2	3.7	3.1	3.7	3.9	5.4	5.2	4.1	3.3	2.8	3.1
	PrecMax	65.0	9.9	8.2	10.4	7.2	10.1	9.7	14.4	14.3	13.8	10.1	5.7	6.2
	PrecMin	26.5	1.1	0.6	0.5	0.4	0.6	1.1	0.4	0.7	0.3	0.0	0.1	0.7
	PrecStd	8.3	1.6	1.5	1.9	1.6	1.7	2.0	3.1	3.2	3.1	2.0	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	75.8	80.7	87.4	91.1	95.4	97.5	95.0	91.1	85.4	77.9	75.3
		MCWB	63.3	64.0	64.2	67.8	72.9	76.3	77.9	77.9	74.9	71.9	66.3	65.1
	2%	DB	67.8	69.5	75.5	83.0	87.4	92.1	94.1	91.8	87.4	81.8	73.4	69.4
		MCWB	61.1	58.9	61.6	66.7	71.6	75.4	77.5	76.9	74.5	71.0	64.8	63.2
	5%	DB	63.4	64.8	70.7	78.9	84.1	89.4	91.6	89.4	84.3	78.2	69.7	65.5
		MCWB	58.1	57.3	59.7	65.0	70.5	74.6	76.7	76.2	73.4	69.3	62.7	60.8
	10%	DB	58.5	59.9	66.0	74.2	80.3	86.4	89.0	86.8	81.5	74.8	66.1	61.2
		MCWB	53.7	52.8	57.7	63.1	68.6	73.7	76.2	75.4	72.4	67.5	60.3	56.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	65.8	67.2	71.9	76.4	79.4	81.1	80.5	78.1	76.3	70.0	67.9
		MCDB	69.7	71.6	75.1	81.4	85.5	87.8	91.1	89.2	84.7	81.5	73.8	72.4
	2%	WB	62.3	62.3	64.6	69.1	73.9	77.8	79.6	79.0	76.8	73.6	66.9	64.5
		MCDB	66.5	67.9	71.8	78.4	82.6	87.3	89.4	87.3	82.8	78.3	71.3	68.4
	5%	WB	58.7	58.3	62.1	67.1	72.1	76.5	78.4	78.0	75.7	71.5	64.1	61.5
		MCDB	63.0	63.6	69.1	75.8	80.5	85.6	87.8	85.8	81.1	76.2	68.6	65.0
	10%	WB	53.9	53.2	58.7	64.7	70.5	75.4	77.5	77.0	74.6	68.9	61.1	57.1
		MCDB	58.0	58.5	65.2	72.7	78.5	83.7	86.5	84.2	79.5	73.6	65.4	60.8
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.9	16.3	17.5	15.9	14.8	14.2	13.0	12.3	14.7	15.3	14.8
		MCDBR	22.8	24.9	24.9	23.7	20.9	19.3	18.1	17.0	16.3	18.2	20.6	20.1
		MCWBR	17.6	17.3	14.7	11.4	9.3	7.2	6.2	6.3	7.0	9.1	12.7	15.6
	5% WB	MCDBR	21.1	22.7	21.6	20.8	17.9	16.4	15.8	14.2	12.8	14.2	16.8	19.0
		MCWBR	18.1	18.4	15.6	12.2	8.9	6.9	6.7	6.1	6.3	8.4	13.2	16.2
Clear-Sky Solar Irradiance	taub	0.309	0.321	0.348	0.391	0.434	0.482	0.510	0.503	0.407	0.356	0.325	0.314	
	taud	2.541	2.503	2.460	2.351	2.260	2.177	2.115	2.119	2.380	2.515	2.551	2.559	
	Ebn at Noon	282	291	292	283	271	257	248	247	268	274	273	272	
	Edh at Noon	25	29	33	39	43	47	49	49	36	28	24	23	
All-Sky Solar Radiation	RadAvg	741	984	1316	1678	1889	2027	1932	1742	1421	1126	838	661	
	RadStd	60	104	103	120	161	130	94	101	113	125	67	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+194	
Regional (54 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+299	+189	

Nomenclature: See separate page